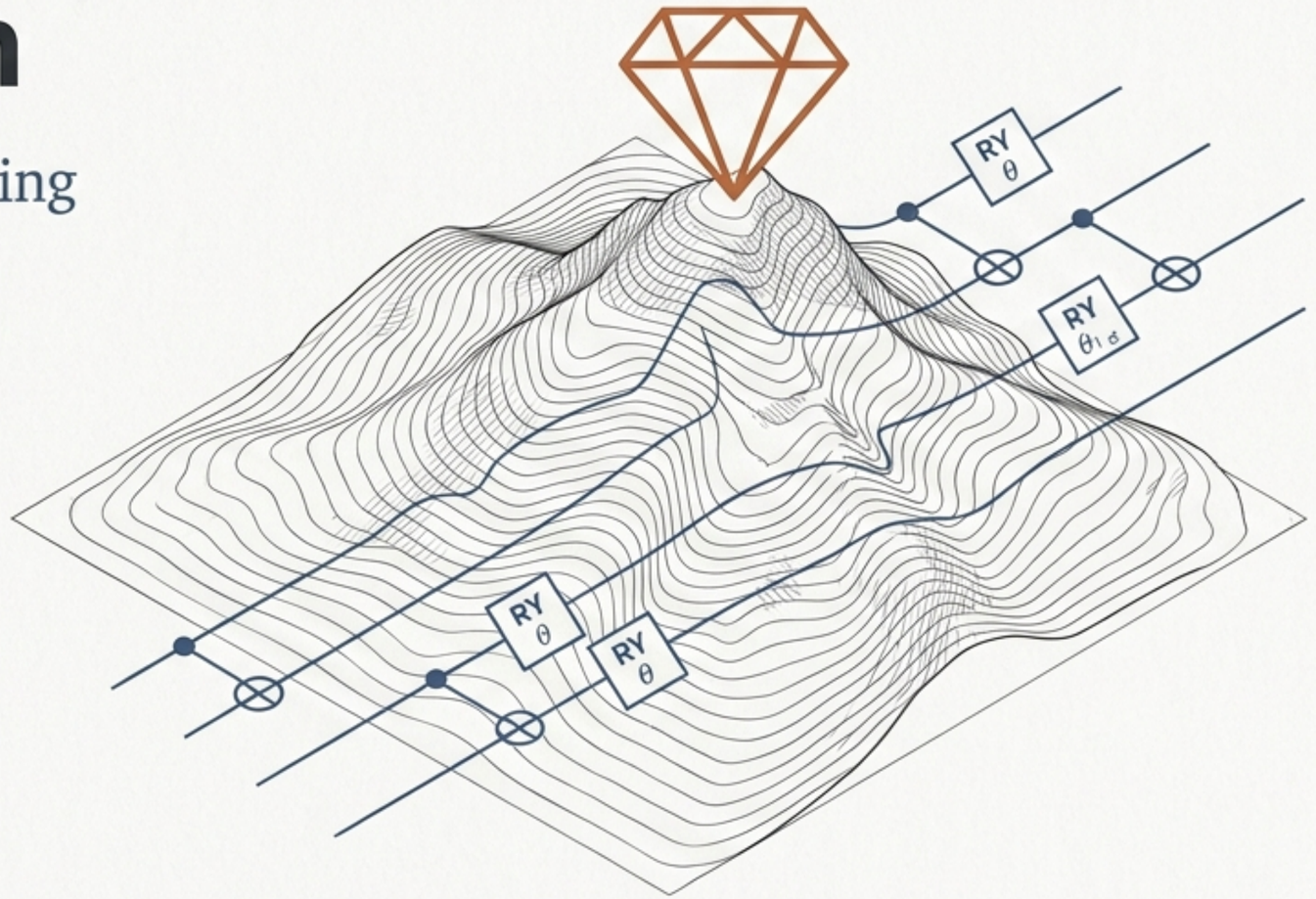
CybraneX: Quantum-Enhanced Mineral Detection

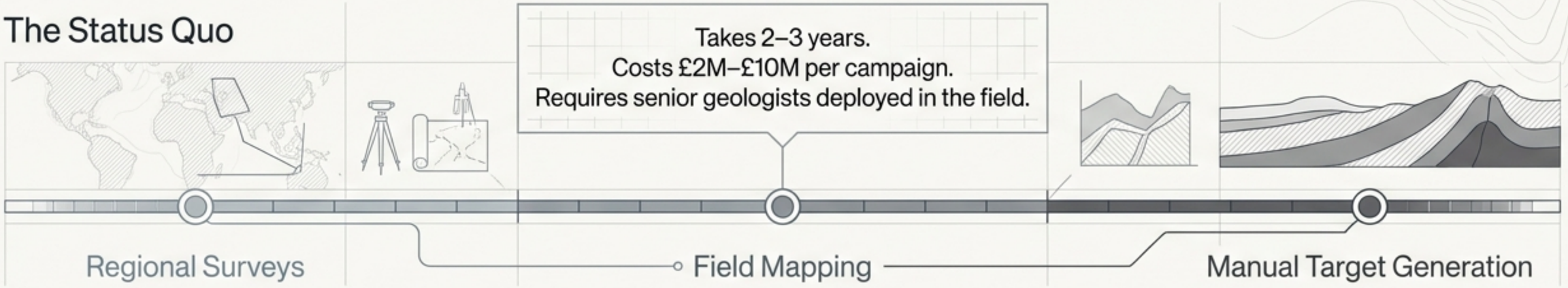
Bridging Commercial Satellite Screening with Quantum Machine Learning.



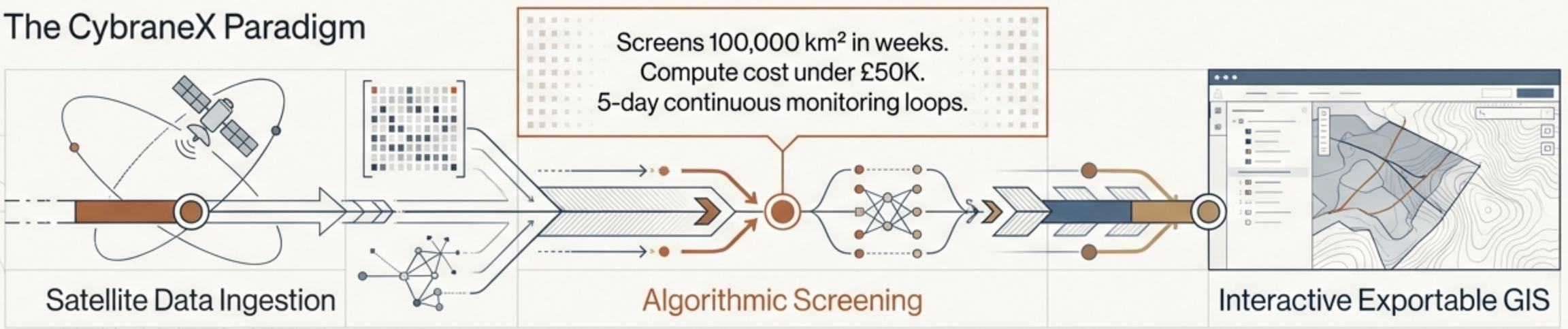
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Eliminating the Exploration Bottleneck

The Status Quo



The CybraneX Paradigm

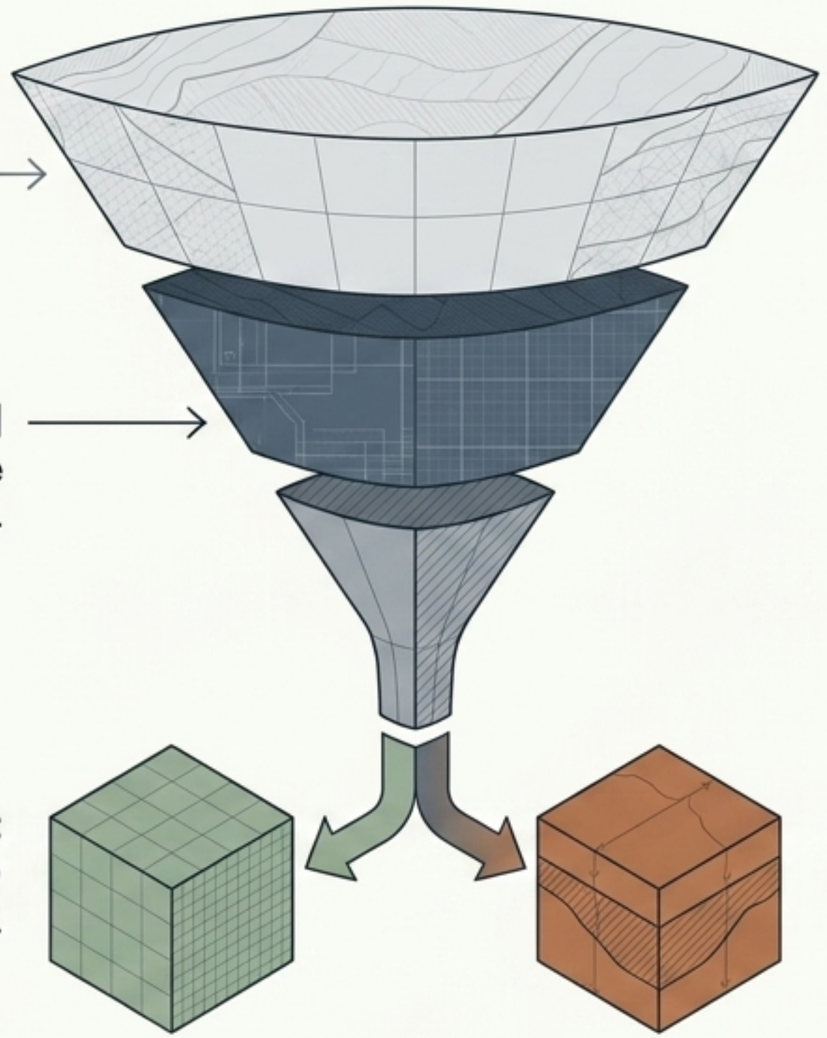


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The Economics of Algorithmic Prospecting

Massive Land Area:
Regional Tenements

Prioritising Recall
We accept a ~41% false-positive rate to guarantee zero missed deposits.



The Filter:
Cheap Soil Sampling (£5,000) eliminates algorithmic false alarms.

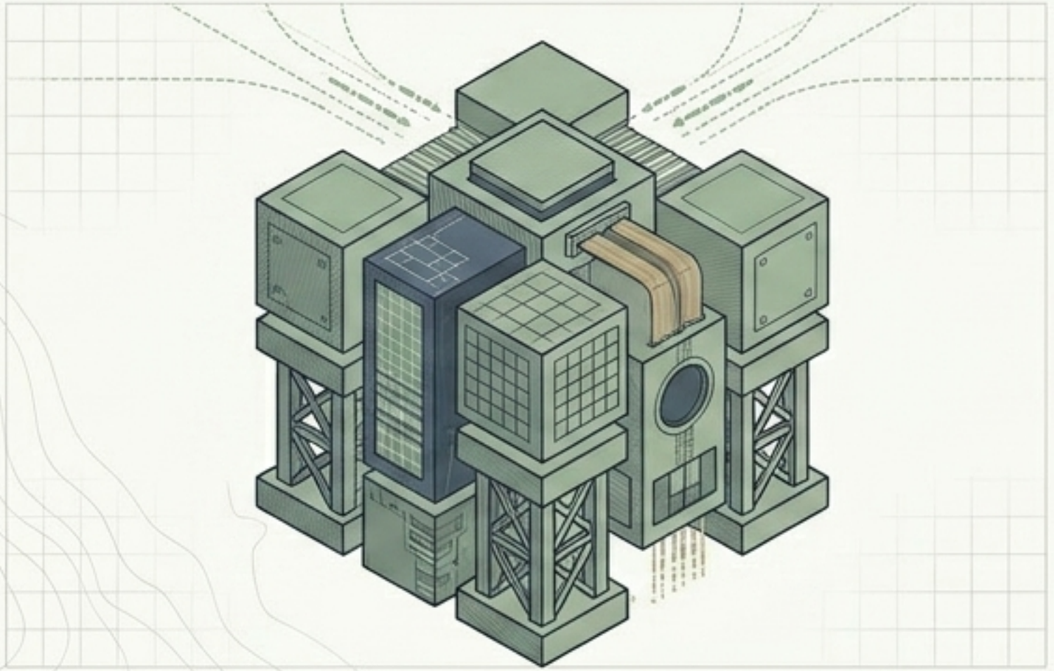
The Prize:
Expensive Drilling (£50M).

We don’t replace drilling; we eliminate 60–80% of wasted spend by telling you exactly where to start.

NotebookLM

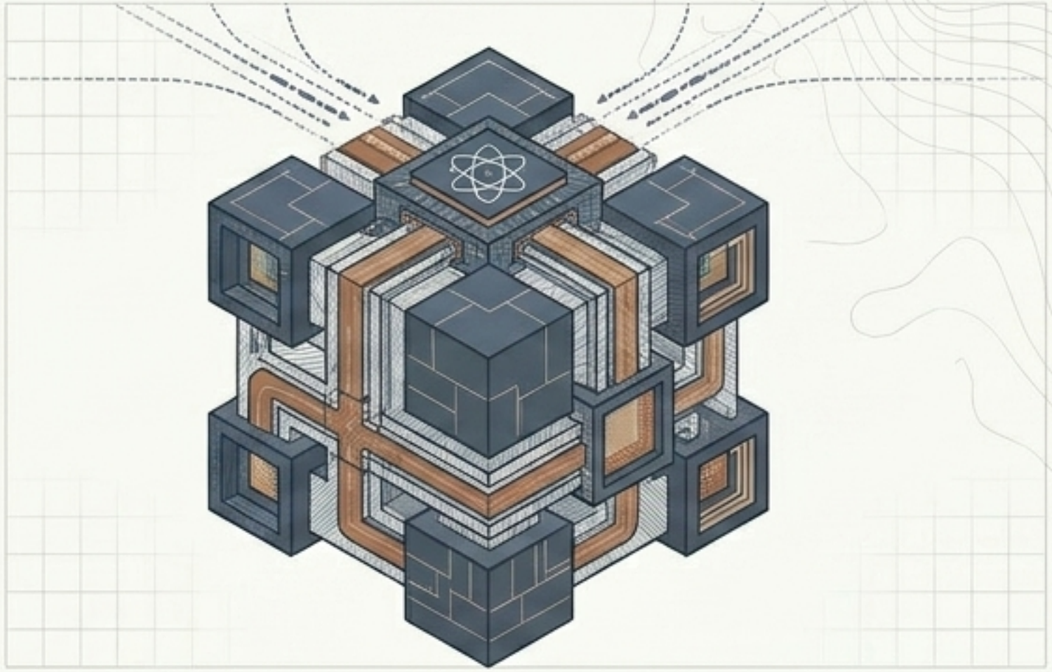
A Dual-Engine Architecture

The Production Layer



- **Input:** Sentinel-2 Satellite Data.
- **Function:** Active commercial screening system. **Highly scalable**, simple B11/B12 clay ratio indices.
- **Status:** Generating Immediate ROI.

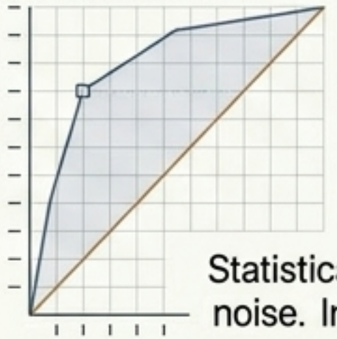
The Innovation Layer



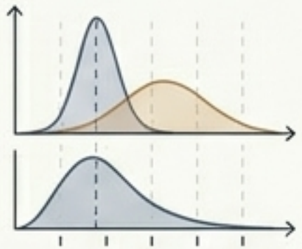
- **Input:** NASA Hyperion Hyperspectral Data.
- **Function:** Quantum Machine Learning research. 198-band feature extraction, complex mineral signatures.
- **Status:** Securing Future IP & Deep Tech Moat.

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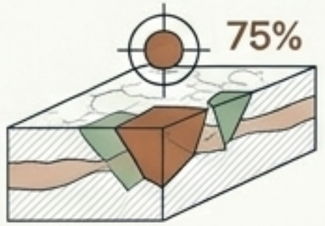
Production System Performance: Proven in the Field



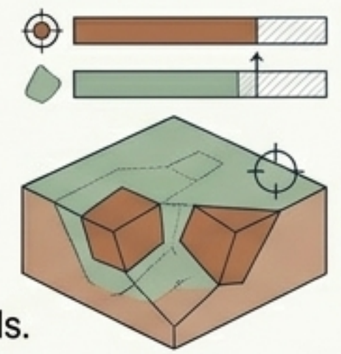
0.82
ROC-AUC



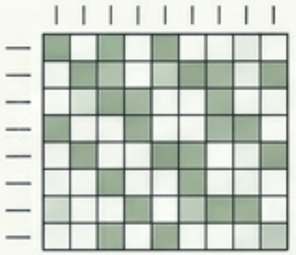
Statistically significant separation from background noise. Independent Mann-Whitney p-value < 0.001.



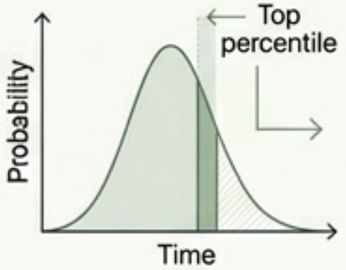
75%
Top-Tier Hit Rate

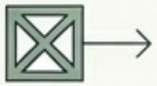


3 out of 4 known rock-hosted deposits appear in the top 5% most prospective pixels.



98th
Percentile Median

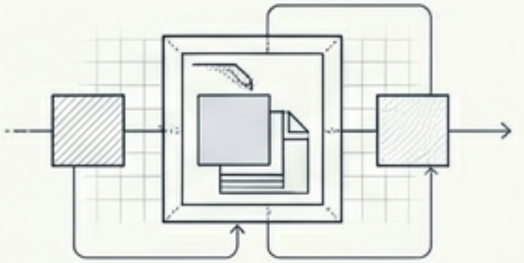




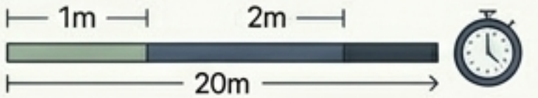
The average known deposit sits in the top 2% of the generated prospectivity map.



Sub-5
Minute Speed



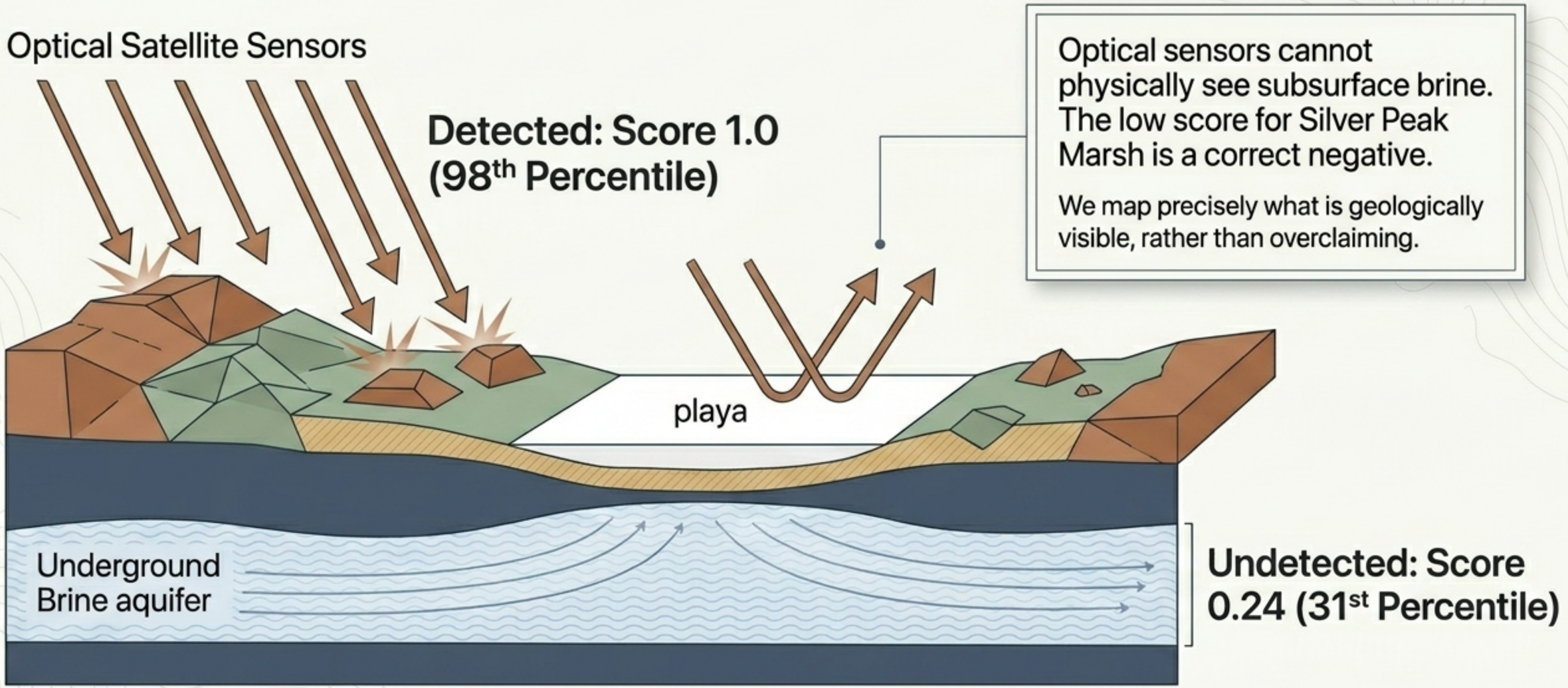
Processes a 10x10 km area at 20m resolution in minutes.



Validated using independent USGS MRDS ground-truth data in Clayton Valley, Nevada.

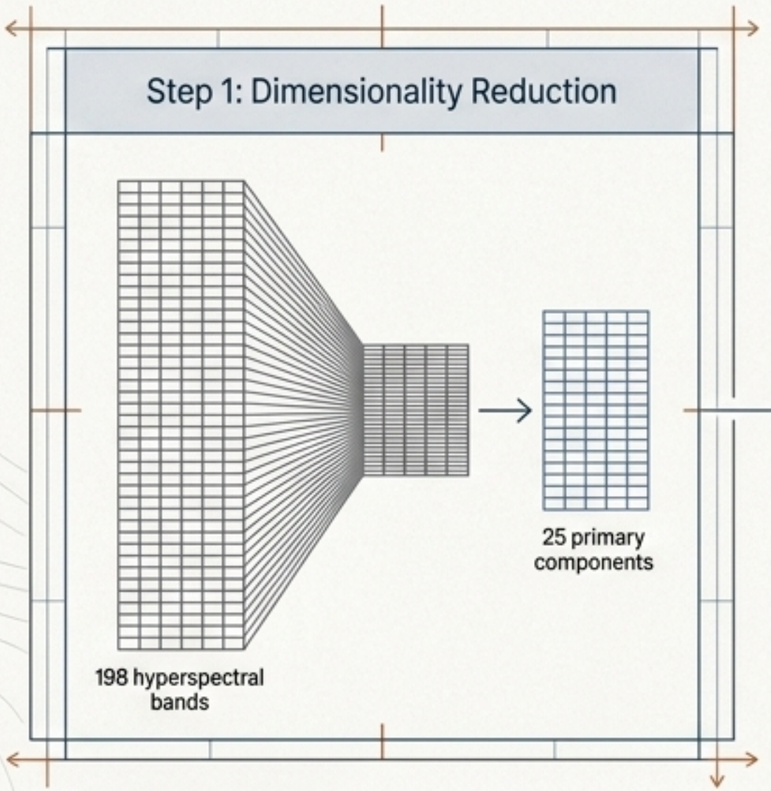
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The Honest Blind Spot: Understanding Sensor Physics



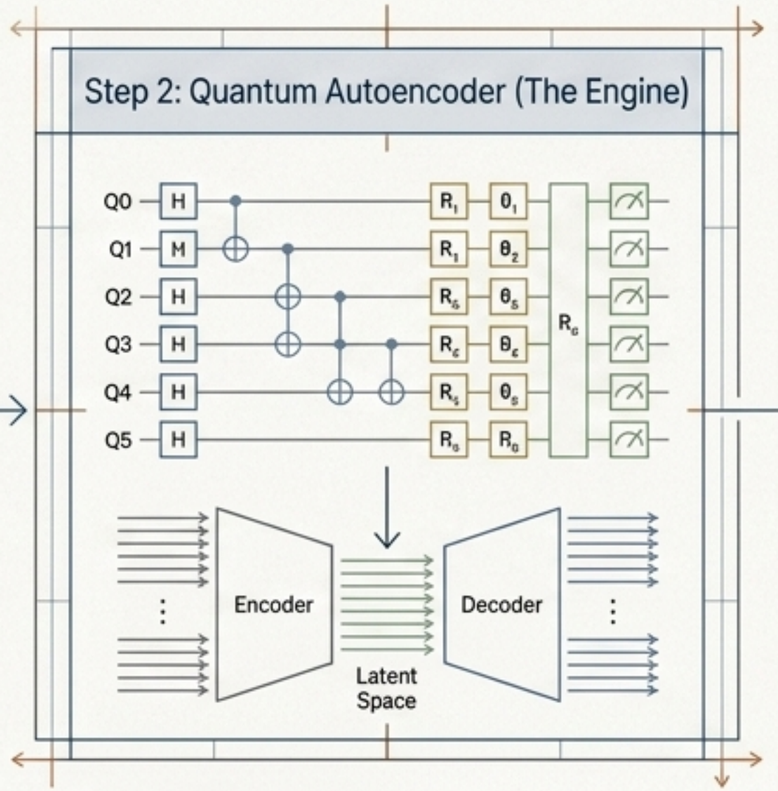
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The Innovation Layer: A 3-Stage Quantum Pipeline



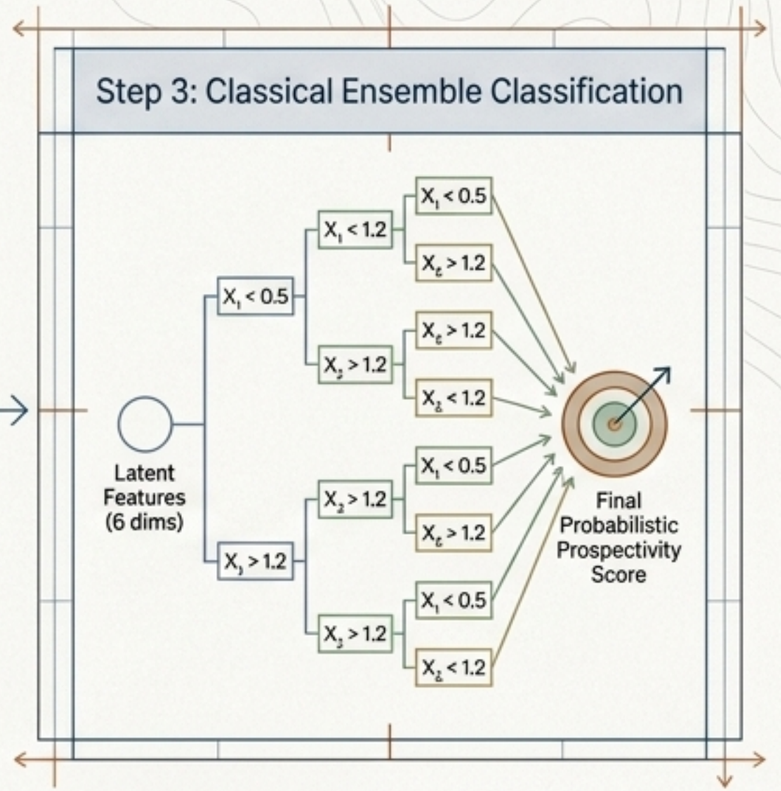
RobustScaler + PCA

Compresses 198 hyperspectral bands down to 25 primary components, retaining 87% of variance.



Non-linear compression

Squeezes the 25 dimensions into a 6-dimensional latent space using PennyLane simulation.



Soft-voting ensemble

(Random Forest, Gradient Boosting, SVM-RBF) provides the final probabilistic prospectivity score.

NotebookLM

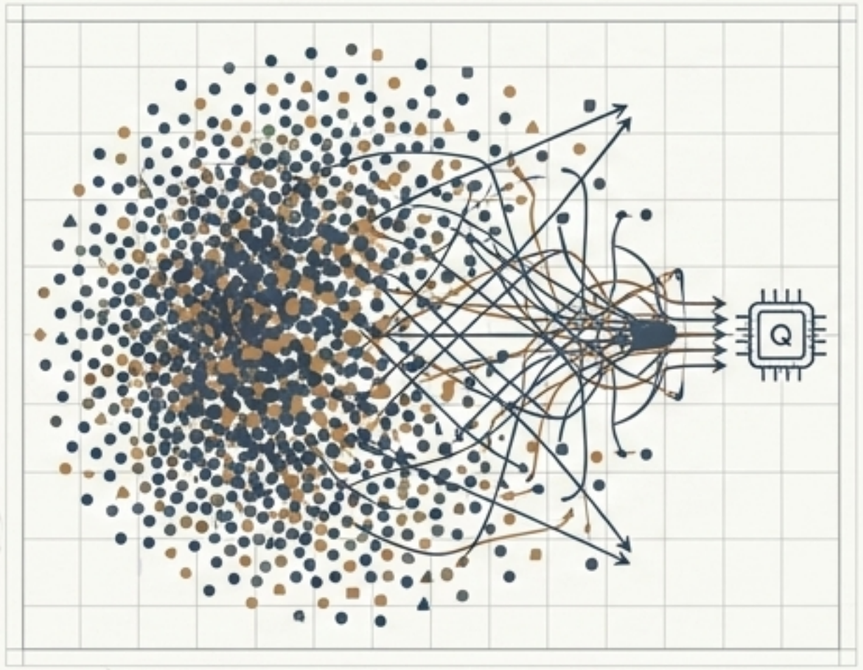
Architectural Showdown: Classical vs. Quantum

Method	Architecture	Role of Quantum	ROC-AUC	Verdict
Method A	PCA Only	None	0.749	Baseline
Method B	PCA + Classical Autoencoder	None	0.762	Strong Classical
Method C	**PCA + Quantum Autoencoder**	**Feature Extractor**	**0.770**	**WINNER (+0.8% AUC Advantage)**
Method D	QSVM Direct	Direct Classifier	0.513	Failed (Too High-Dim)
Methods C2/C3	QAE + QSVM/VQC	Classifier	~0.500	Unstable / Failed
Method F	QAA Hybrid	Adversarial	0.687	Not Competitive

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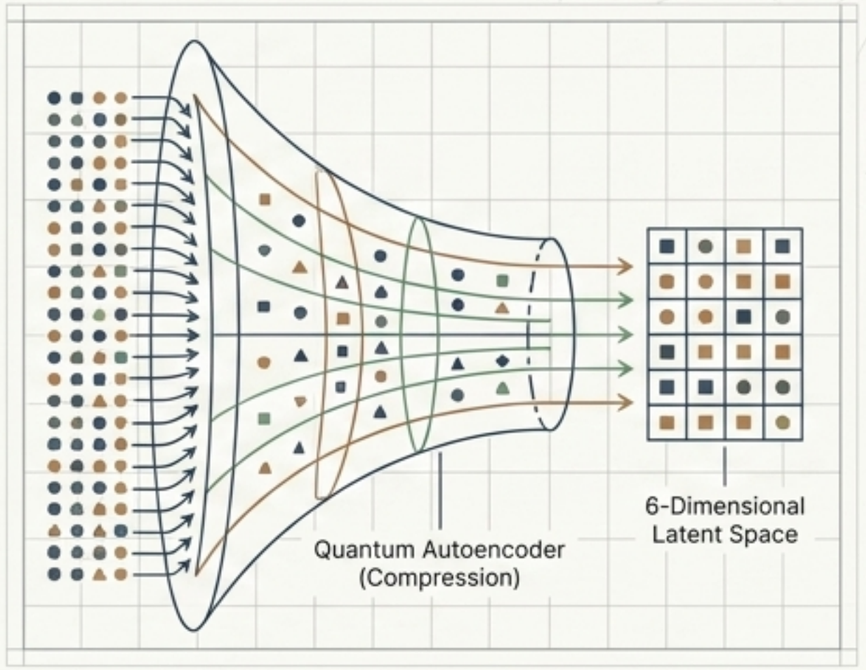
The Core Scientific Insight: Feature Extraction

Quantum as Final Classifier



Direct quantum classifiers (QSVM, VQC) **fail.** Current 6-qubit scales simply cannot handle high-dimensional geospatial datasets, resulting in near-random performance.

Quantum as Feature Extractor

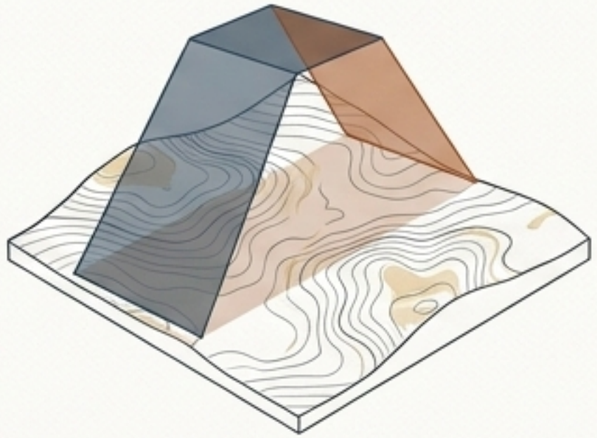


Hybrid models succeed. By using a Quantum Autoencoder strictly for non-linear compression (Method C), we achieve a measurable +0.8% AUC advantage over purely classical autoencoders.

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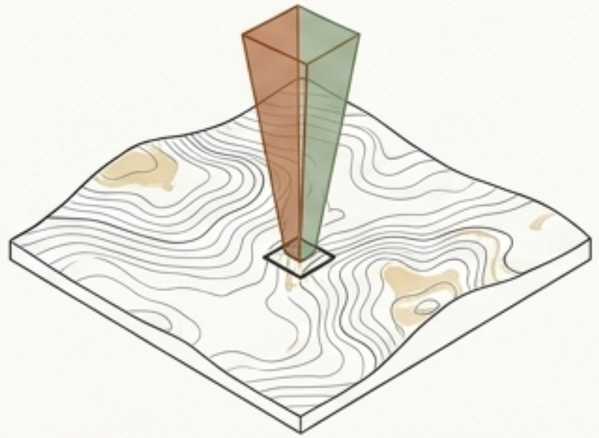
The Geospatial Data Bottleneck

Sentinel-2 (Current Production)



- Only 2 SWIR bands (cannot resolve specific minerals like Li-OH vs Mg-OH).
- 20m resolution misses outcrops under 400m².
- Severe cloud contamination limits (monsoon regions lose 60% of dates).

NASA Hyperion (Research Dataset)



- Instrument retired in 2017 (no new data).
- Extremely narrow 7.7km swath (missed Silver Peak deposit entirely).
- Circular training labels restrict true real-world validation.

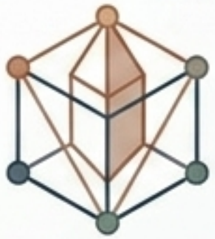
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The Next-Generation Data Stack

The Missing Data	Geological Imperative	Future Satellite / Tech Source
Continuous SWIR (2000-2500nm)	Distinguish exact Li-bearing clays (hectorite vs generic Al-OH)	NASA EMIT, ASI PRISMA, DLR EnMAP
Thermal Infrared (8-12µm)	Identify quartz and carbonate minerals lacking VNIR/SWIR signatures	ASTER TIR, Landsat TIRS, future HyTES
Subsurface Sensing	Detect hidden brine aquifers beneath salt flats	Airborne Electromagnetics (AEM), Gravity Surveys
<5m Spatial Resolution	Resolve individual, small-scale alteration zones	Commercial WorldView-3 SWIR licensing

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Core Mining & Exploration Markets



Lithium ($22B EV Supply Chain)

Targeting hard-rock pegmatites, clay alteration (hectorite, montmorillonite), and surface evaporites.



Copper-Gold ($180B/year)

Mapping phyllic and argillic alteration haloes (sericite, kaolinite) in complex porphyry systems.



Vanadium (Grid Storage Batteries)

Roscoelite (V-bearing muscovite) detection within extensive carbonaceous shales.



Nickel Laterite (EV Cathodes)

Mapping precise iron-oxide and serpentine mineral aggregations.

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Value Delivered: Government vs. Enterprise

Government Agencies (USGS, GSI, CSIRO)

Focus: Speed & Scale.

- Collapse 3-year regional geological surveys into weeks.
- Replace £10M traditional mapping campaigns with £50K algorithmic screening.
- Output interactive GIS files, ending reliance on locked PDFs.

Exploration Companies

Focus: Risk Reduction & Ranking.

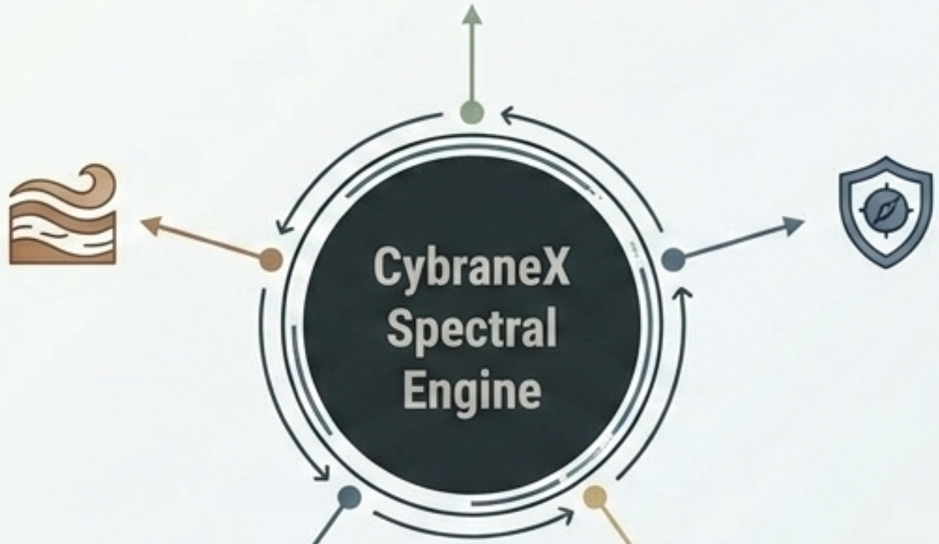
- Ranks 100% of a given tenement by prospectivity with explicit uncertainty bounds.
- Delivers actionable initial results in 48 hours.
- Focuses expensive drilling budgets strictly on mathematically proven target zones.

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Expanding the Horizon: Beyond Mining

Agriculture: Soil mineral mapping (potassium, phosphorus) for precision fertilisation.

Water Resources: Aquifer characterisation via evaporite and clay mapping in arid zones.



Defence & Intelligence: Critical mineral supply chain mapping in contested global regions.

Environmental: Acid mine drainage detection and contamination plume mapping.

Carbon Sequestration: Locating ultramafic rocks (olivine) for optimal mineral carbonation (CCUS) sites.

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The Future of Planetary Sensing

Production-Ready Today

An active Sentinel-2 pipeline delivering 0.82 AUC commercial mineral targeting immediately.

Quantum-Secured Tomorrow

Proprietary Quantum Autoencoder architectures proving early-stage non-linear feature advantage.

Infinitely Scalable

A core algorithmic engine ready to ingest next-generation hyperspectral data for global cross-industry application.

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